MATHEMATICS · NUMBER · FOUNDATION

Number Bases

Base 10, Base 2(binary), Base 8(octal), Base 16(hex) — and how to convert between them.

Use these notes well and you will be able to:

- Explain what a number base means and why base 10 uses digits 0–9.
- Convert decimal (base 10) numbers to binary, octal, and hexadecimal.
- Convert binary, octal, and hex numbers back to base 10.
- Add and subtract binary numbers without converting to base 10.
- Recognise and apply the relationship: 1 hex digit = 4 binary bits.

If you follow the steps, you will succeed.

What Is a Number Base?

Positional Value — How Any Base Works

Base10— positional values

1000	100	10	1
10^3	10^2	10^1	10

position

Uses digits: 0–9

Example: (11) = 1×10 + 1×1 = 11

Base 2—positional values

8	4	2	1
2^3	2^2	2^1	2

position

Uses digits: 0–1

Example: (1011) = 1×8 + 0×4 + 1×2 + 1×1 = 11

Key Words

KEY WORD	DEFINITION
Base / Radix	The number of unique digits used. Base 10 uses 0–9. Base 2 uses 0–1.
Binary (base 2)	Uses only 0 and 1. Each position is a power of 2. Used in computers.
Octal (base 8)	Uses 0–7. Each position is a power of 8. Compact form of binary.
Hexadecimal (base 16)	Uses 0–9 then A–F (A=10, B=11, ..., F=15). 1 hex digit = 4 binary bits.
Denary / Decimal	Base 10 — our everyday number system using digits 0–9.
Bit	A single binary digit — either 0 or 1.
Nibble	4 bits. One hex digit represents one nibble.
Byte	8 bits. Two hex digits represent one byte.
Subscript notation	(101) means 101 in base 2. (3F) means 3F in base 16.

Converting: Decimal Other Bases

METHOD: REPEATED DIVISION

Divide by the base repeatedly. Read remainders BOTTOM to TOP.

Repeated Division — Decimal to Binary, Octal, Hex

25 Base 2

$25 \div 2 = 12$	r 1
$12 \div 2 = 6$	r 0
$6 \div 2 = 3$	r 0
$3 \div 2 = 1$	r 1
$1 \div 2$	r 1

read up

25 = 11001

45 Base 8

$45 \div 8 = 5$	r 5
$5 \div 8 = 0$	r 5

read up

(55)

173 Base 16

$173 \div 16 = 10$	r D
$10 \div 16 = 0$	r A

read up

(AD)
A=10, D=13

Converting: Other Bases Decimal

METHOD: EXPAND BY POSITIONAL VALUE

Multiply each digit by positional value (baseposition), then add.

Convert (11001) to decimal

Positions (right to left): 0, 1, 2, 3, 4

$= 1\times2 \ + 1\times2^3 + 0\times2^2 + 0\times2^1 + 1\times2$

$= 16 + 8 + 0 + 0 + 1$

= 25

Convert (2B7) to decimal (B = 11)

$= 2\times16^2 + B\times16^1 + 7\times16$

$= 2\times256 + 11\times16 + 7\times1$

$= 512 + 176 + 7$

= 695

Binary Arithmetic

ADD

10 in binary = 0 carry 1

SUB

Borrow 10 from next column

KEY

Always carry when sum ≥ 2

Add in binary: 1011 + 1110

1 0 1 1
+ 1 1 1 0

Col 0 (right): 1+0 = 1
Col 1: 1+1 = 10 write 0, carry 1
Col 2: 0+1+1(carry) = 10 write 0, carry 1
Col 3: 1+1+1(carry) = 11 write 1, carry 1
Col 4: 0+0+1(carry) = 1
=1 1 0 0 1 Check:11+14 =25

Binary Hex Shortcut

1 HEX DIGIT = 4 BINARY BITS (nibble)

Group binary into fours from right, convert each group independently.

This works because 16 = 2. Example: (10110111)   (B7)

Hex	0	1	2	3	4	5	6
Bin	0000	0001	0010	0011	0100	0101	0110
Hex	**7**	**8**	**9**	**A**	**B**	**C**	**D**
Bin	0111	1000	1001	1010	1011	1100	1101
Hex	**E**	**F**					
Bin	1110	1111					

Convert (10110111) to hex using the nibble method

Group into 4s from right: 1011 | 0111
1011 = 8+2+1 = 11 = B
0111 = 4+2+1 = 7 = 7
(10110111) = (B7)

Practice Questions

DECIMAL BINARY / OCTAL / HEX

1. Convert 42 to binaryusingrepeated division.

2. Convert 100 to octal.

3. Convert 255 to hexadecimal.

BINARY / OCTAL / HEX DECIMAL

4. Convert (101101) todecimal.

5. Convert (73) to decimal.

6. Convert (3C) to decimal.

ADD YOUR NOTES HERE

Hey smart student!

Every base usesthe SAME idea: positional value × digit, then add.

Binary to hex: group 4 bits = 1 hex digit. Always from the right.

Repeated division: divide by base, read remainders BOTTOM to TOP.

Keep going — see you in the next lesson! From FREDi